

S1 & S2 — IVY MD0630 Leakage Sensor: Modbus Bring-up, Register Map & Live ESP32 Read

OpenAMI / MeshEMS — Continuous Leakage Detection (NESL)

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2026-07-22

1 Objective

Sprint **S1** of the leakage-detection assignment (supervisor: Glenn, NESL): get **RS-485 / Modbus serial communication working** — **first PC → IVY RCM CT** — and **derive the Modbus register map** (offsets, function code, scaling) so the ESP32 driver (S2) can read continuous AC/DC leakage.

Result: S1 and S2 are both complete. The module communicates, the full register map has been derived from the live device and confirmed against IVY's own PC tool (**CT.exe**), and the ESP32 driver now reads the live module **reliably over its own RS-485/UART bus** (Section on S2).

2 Bench setup

The module was brought up on the PC first (per Glenn's directive), using a USB-TTL adapter as the Modbus master interface. Because the lab's USB-TTL cable exposes no VCC wire, the required TX/RX pull-ups were fed from an ESP32 5 V pin, with **all grounds tied to a single common ground**.

S1 bench: PC reads the MD0630 through the USB-TTL; the ESP32 5 V feeds the pull-ups

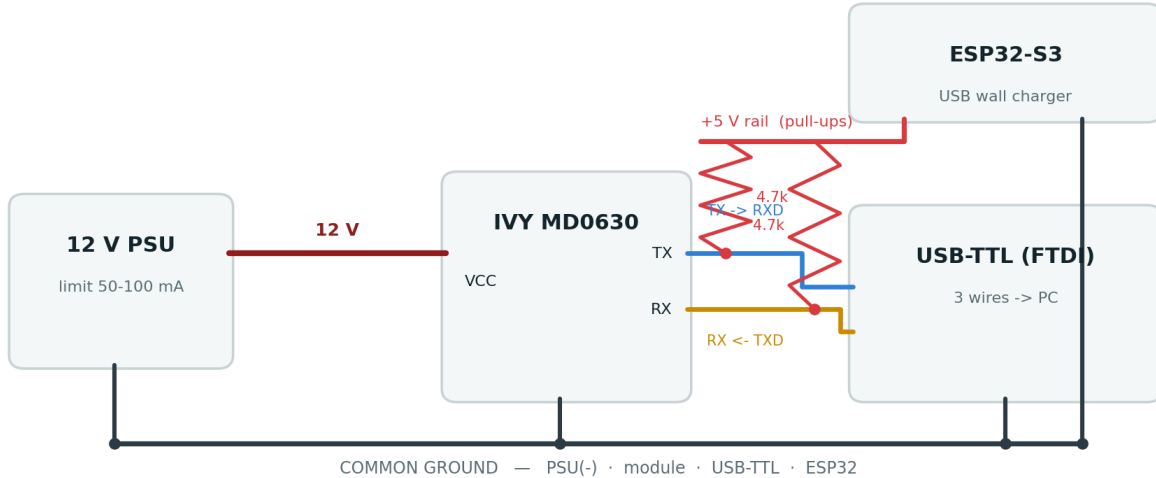


Figure 1: S1 bench wiring — 12 V powers the module; the PC reads it over the USB-TTL; the ESP32 5 V feeds the pull-ups; every ground is common.

Key wiring points (the ones that unblocked communication):

- Module powered at **12 V** (VCC/GND — never reversed/shorted).
- Module **TX** → **USB-TTL RXD**, module **RX** → **USB-TTL TXD** (crossed).
- **4.7 kΩ pull-ups** on TX and RX (the datasheet requires them; the module TX is open-drain).
- **Common ground** across PSU(-), module, USB-TTL **and** the ESP32 that sources the 5 V rail — this last link was essential.

3 Serial parameters (confirmed)

Confirmed from IVY's `CT.exe` “Serial Ports” dialog and reproduced with a direct Modbus prober:

Parameter	Value
Baud / format	9600 · 8 · N · 1
Slave address	1 (<code>CT.exe</code> uses broadcast 0x00 in its frames; addr 1 also answers)
Function code	FC 0x03 (holding) — FC 0x04 (input) returns the same values
Scaling	raw × 0.1 mA (60 → 6.0 mA, 300 → 30.0 mA)

4 Register map (derived)

`CT.exe` reads the device in three groups. Decoding its frames and matching them to the tool's labelled fields gives the following map.

Table 1: Derived MD0630 register map. Defaults match the datasheet (DC 6 mA, AC 30 mA).

Register	Raw (default)	$\times 0.1$ mA	Field	Access
0x0000	0	0.00	DC Leakage Current	R
0x0001	0	0.00	AC Leakage Current	R
0x0002	60 (0x003C)	6.0	DC Threshold	R/W
0x0003	300 (0x012C)	30.0	AC Threshold	R/W
0x0100	1	—	Version Number	R
0x0111	5	—	Minor Version Number	R
Modbus ID	1	—	slave address	R/W

Note. The DC-vs-AC assignment of 0x0000/0x0001 is inferred from the DC-before-AC ordering of the thresholds (both leakage registers read 0 at rest); it should be confirmed by inducing a known DC or AC leak and re-reading.

5 Evidence — CT.exe communication log

The raw Modbus frames captured from IVY’s tool (authoritative):

```

Write : 00 03 00 00 00 06 C4 19
Read  : 00 03 0C 00 00 00 00 00 3C 01 2C 01 2D 00 00 AF 91    -> 0, 0, 60, 300, 301, 0
Write : 00 03 01 00 00 01 84 27
Read  : 00 03 02 00 01 44 44                                -> 0x0100 = 1    (Version)
Write : 00 03 01 10 00 02 C5 E3
Read  : 00 03 04 00 00 00 05 7B 30                            -> 0x0111 = 5    (Minor Version)

```

CT.exe labelled read-out: Modbus ID 001 · Version 0001 · Minor Version 0005 · DC Threshold 6.0 mA · AC Threshold 30.0 mA · DC Leakage 0.00 mA · AC Leakage 0.00 mA.

A second, independent read using a direct prober (address 1, our own frames) returned the same block:

```

TX : 01 03 00 00 00 06 C5 C8
RX : 01 03 0C 00 00 00 00 00 3C 01 2C 01 2D 00 00 ...

```

6 Method summary

1. Wire PC module over the USB-TTL with pull-ups and a shared common ground.
2. Confirm the link: the module answers at address 1, 9600 8N1 (both FC03 and FC04).
3. A read of 0x0000+7 returned exception 0x02 (illegal data address) → the block is not 7-wide.
4. Scan single registers → valid block 0x0000–0x0005; a 6-register read returns a clean frame.
5. Cross-check with CT.exe: its three read groups and labelled fields fix every register’s meaning.

7 Impact on the design

- **Answers the register-map question.** The real layout is **neither** the earlier AI-proposed spec (AC leakage at 0x00, thresholds at 0x04/0x05) **nor** the raw CT.db field order — it is
 - 1.
- **Driver updated.** `include/metering/modbus_md0630.h` `RegMap` and `read_fc = FC_HOLDING` now carry these confirmed offsets; the mock pipeline (data model → MQTT subpanel_RCMleaks) is unchanged.

8 S2 — Live read on the ESP32

With the register map known, the module was moved from the PC USB-TTL onto the **ESP32’s own RS-485/UART bus** and the `Modbus_MD0630` driver was switched from mock to **real reads**.

8.1 Wiring (ESP32)

Module	ESP32	Note
TX	GPIO42 (RS485_1 RX)	+ 4.7 kΩ pull-up to 3V3
RX	GPIO7 (RS485_1 TX)	+ 4.7 kΩ pull-up to 3V3
GND	GND	common ground (ESP32 + module + 12 V PSU—)
VCC	12 V PSU	never the ESP32

The MD0630 is UART, so it connects **directly** to the ESP32 UART pins (no RS-485 transceiver). Pull-ups go to **3V3** — the ESP32-S3 GPIOs are **not** 5 V tolerant. The pins are the **GPIO numbers 42 / 7**, not the silkscreen “RX”/“TX” (which are the USB console UART, GPIO44/43).

8.2 Evidence — serial capture (module at rest)

```
SETUP: MODBUS: MD0630 leakage CT: address:1
MD0630 [addr:1 fc:0x03]: AC=0.00 mA DC=0.00 mA (ok:9 fail:0)
MD0630 [addr:1 fc:0x03]: AC=0.00 mA DC=0.00 mA (ok:10 fail:0)
MD0630 [addr:1 fc:0x03]: AC=0.00 mA DC=0.00 mA (ok:11 fail:0)
MD0630 [addr:1 fc:0x03]: AC=0.00 mA DC=0.00 mA (ok:12 fail:0)
```

12 consecutive successful reads, 0 failures. The driver reads 0x0000/0x0001 at FC 0x03, scales ×0.1 mA, and feeds `leakageModel` (published on `subpanel_RCMleaks`).

8.3 Two gotchas resolved

- **Wrong pins:** the module was first on the silkscreen “RX/TX” (USB console, GPIO44/43) instead of **GPIO42/GPIO7** → timeouts; moving to the numbered pins gave the first successful read.
- **Address collision:** the SHT20 is polled at address 1 — the same as the CT during bring-up → intermittent reads. A **LEAKAGE_BRINGUP** build flag skips the SHT20 poll → reliable reads. In production the CT is commissioned to node 100, so both coexist.

9 Next steps (S3)

1. **S3 — threshold writes:** write 0x0002 (DC) / 0x0003 (AC) via **FC 0x06**, value = mA × 10. **CT.exe** shows an **unLock ID** step, so derive/verify the unlock sequence first (by sniffing a **CT.exe** write), and **always read back to confirm** (per Glenn). Default life-safety values: **DC 6 mA**, **AC 30 mA**.
2. **Confirm DC-vs-AC** of 0x0000/0x0001 by inducing a known leak.
3. **Production:** commission the CT to node **100**, remove **LEAKAGE_BRINGUP**, re-enable WiFi/MQTT.

Repository: docs/leakage_MD0630TA1A/ — full details in **S1_register_map_findings.md**; driver in **include/metering/modbus_md0630.h** + **src/metering/modbus_md0630.cpp**.